Topic:

Managing Control files & Redo Log files:

⇒ An Oracle database consists of three types of mandatory files:

- datafiles
- Control files
- online redo logs

⇒ Control files and redo logs are critical database files.

⇒ These files are critical because a normally operating database cannot function without these files.

⇒ Control files are small binary files that contain information regarding the structure of the database.

⇒ Online redo logs are crucial files that record the transaction history of the database.

Control Files

- Definition and content is on previous page.

Displaying the Contents of a Control file.

- Displaying the content of control files directly is not typically possible since control files are binary files containing vital database metadata.
- However, much of the information stored in control files can be obtained through various dynamic performance views in oracle.

1. Control file information:

⇒ To get the names and locations of the control files:

- SELECT name FROM v$controlfile;

2. Database Structure information:

⇒ To list all data files:

- SELECT file name FROM dba_data files;

3. redo log files:

⇒ To list all redo log files:

- SELECT member FROM v$ logfile;

4 Tablespaces:

⇒ To list all tablespaces:

- SELECT tablespace-name FROM dba-tablespace;

5. Checkpoint information:

⇒ To display checkpoint information:

- SELECT checkpoint_change# FROM V$database;

6. Backup information:

⇒ SELECT * FROM V$backup;

7. Control file record sections:

⇒ To get information about different section within the control files:

- SELECT Type, record_size, records_total, records_used FROM V$controlfile_record_section;

2. Adding a control file:

1. Alter the initialization file CONTROL_FILES parameter to include the new location and name of the control file.
2. Shut down database.
3. Use an OS command to copy an existing control file to the new location and name.
4. Restart db.

3. Moving a control file:

1. Determine the CONTROL_FILES parameter's current value.
2. Alter CONTROL_FILES parameter to reflect that you're moving a control file.
3. Shut down database.
4. At the OS prompt, move the control file to the new location.
5. Start up database.

4. Removing a control file:

1. Identify which control file has experienced media failure by inspecting the alert.log for information.
2. Remove the unavailable control file name from the CONTROL_FILES parameter.
3. If using an init.ora file, modify the file directly with an OS editor. If using an spfile, modify the CONTROL_FILES parameter with the ALTER SYSTEM statement.
4. Stop and start db.